

claude-windows / 7fe4ef45-d30a-448e-98eb-4da4fba244f1

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\7fe4ef45-d30a-448e-98eb-4da4fba244f1\subagents\workflows\wf\_b531eaeef-39c\agent-aa48a629bf8426ae1.jsonl`
- SHA-256: `5a78c03f5230e0ff55f08409417cb30963114d7d7be4f00f32a1c737606f921c`
- Source modified: `2026-06-19T17:10:12+00:00`
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- project: `wf\_b531eaeef-39c`
- session_id: `7fe4ef45-d30a-448e-98eb-4da4fba244f1`

Transcript

1. user (2026-06-19T16:59:09.585Z)

You are an adversarial design reviewer. Be skeptical; the goal is to find real flaws BEFORE we build.

CONTEXT ? badminton/racket-sports rally detector. Pipeline: WASB shuttle DETECTOR (heatmap ? per-frame candidate list) ? single-target online TRACKER (collapses to ONE (x,y) per frame) ? velocity windower ? rallies.

MEASURED (GX010142, owner's anchor clip, on the 21-25s window, from the cached det_raw.jsonl detector candidates):

- WASB's detector emits ≥ 2 candidates on 31% of detected frames (cached, never used past the tracker).
- During the 21-25s rally: 93% of frames carry the ADJACENT-court shuttle (x-norm ~ 0.88 , what the tracker followed),
 - 83% ALSO carry a PROMINENT-court shuttle (x-norm ~ 0.56), 76% carry BOTH at once.
- A greedy nearest-neighbour reconstruction of the prominent-court candidates yields a CONTINUOUS rally arc
 - (75% window coverage, 431px vertical excursion, 8.6px median per-frame jump, ~ 2.4 s sustained)

? VISUALLY CONFIRMED

as a real foreground-court rally (player mid-stroke ? net ? second player), which the single-target tracker discarded.

PROPOSED DESIGN ("the Which problem"):

Split the tracker's two conflated jobs:

- Layer 1 "see everything": a MULTI-TARGET linker over the ALREADY-CACHED det_raw.jsonl candidates ? a SET of shuttle
 - tracks per region (top-K), each with trajectory + duration + motion-energy. Pure-Python over the cache, ZERO GPU,
 - WASB stays frozen (MIT).
- Layer 2 "pick the match's story": score each track by court-membership (the existing prominent-court polygon +
 - point_in_polygon) + rally-likeness (motion energy, vertical excursion, ≥ 2 net-crossings/Gate-A, ends on shuttle-floor-contact) and SELECT the track(s) whose story is the match's; feed those to the windower.

Claimed benefits: recall-POSITIVE (recovers the discarded prominent rally ? a G1 miss) AND precision-positive (drops the adjacent rally) ? unlike today's court_gate which is a subtractive post-filter on the single track (precision-only). Also claimed to SUBSUME the "ghost" (a stuck-pixel false-lock track has ~ 0 motion energy ? never selected). Lands default-OFF, flag-gated, measured on cached candidates + golden labels with no GPU; single-track tracker stays the default.

YOUR LENS:

Attack the EVAL METHODOLOGY + n=1 OVERFITTING. The whole design is validated on ONE clip/ONE window (GX010142 21-25s). How do we measure ?P/?R/?F1 HONESTLY on cached candidates + golden labels with no GPU, on the dual eval sets (raw + golden), with an over-seg guard, WITHOUT designing the selection heuristics to fit GX010142? What is the minimum corpus/protocol before any default-ON flip? Mitigations + a concrete measurement protocol.

Return concrete, technically-grounded risks with severity + mitigations. Mark kills_design=true ONLY for a fundamental unmitigable flaw.

2. assistant (2026-06-19T16:59:13.402Z)

I'll investigate the actual cached data and code to ground this adversarial review in reality rather than just reasoning abstractly. Let me start by pulling the repo and reading the key docs.

3. user (2026-06-19T16:59:16.900Z)

Your configuration specifies to merge with the ref 'refs/heads/docs/court-floor-rally-end' from the remote, but no such ref was fetched.

---BRANCH---

docs/court-floor-rally-end

---STATUS---

?? artifacts/

?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md

4. user (2026-06-19T16:59:17.082Z)

ABLATION_LAYER.md
ANALYZERS_AND_RECONCILERS.md
BACKLOG.md
CODE_AUDIT_AND_TEST_HARDENING.md
CODE_MAP.md
COMMERCIALIZATION.md
DEFERRED_HARDENING_PLAN.md
DESKTOP_APP_PLAN.md
EVALUATION.md
EXECUTION_POLICY.md
EXPERIMENT_HARNESS.md
FIELD_VISIT_ANSWER_SHEET.md
FIELD_VISIT_PLAYBOOK.md
GOLDEN_DATA_SHARING.md
GOLDEN_REGRESSION_FIXTURES.md
GOLDEN_SET_IMPLEMENTATION.md
GOLDEN_SET_PHASE1_VIDEOS.md
GOLDEN_SET_VENDORING.md
GOLDEN_VIDEOS.md
HARDENING_LOOP_HANDOFF.md
HOSTING_PLAN.md
HOW_RALLY_DETECTION_WORKS.md
I18N_PLAN.md
KHELSUTRA_PRODUCT_OVERVIEW.md
MULTI_SIGNAL_FUSION_PLAN.md
NEXT_STEPS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PERSONALIZATION_PLAN.md
PER_VIDEO_OUTPUT_LAYOUT.md
PLATFORM_ARCHITECTURE.md
PMF_FIELD_SIGNALS.md
PMF_MARKET_RESEARCH.md
POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
PRODUCT_FIELD_ASSOCIATE_FLYER.md
PROJECT_DOC_TEMPLATE.md
PROMINENT_COURT_DETECTION.md

QUALITY_ITERATIONS.md
R1_MILESTONE_LAUNCH_REPORT.md
R2_EVAL_METRIC_DESIGN.md
RALLY_DETECTION_GAP_CLOSURE.md
RALLY_DETECTION_QUALITY_REPORT.md
RALLY_QUALITY_RESEARCH.md
README.md
REAL_FOOTAGE_VALIDATION_REPORT.md
ROADMAP.md
ROORA_LABELING_GUIDELINES.md
STORAGE_SHARING_MODEL.md
TRACKNET_WSL_SETUP.md
UI_ALPHA_EXECUTION_PLAN.md
UI_PROPOSAL.md
VIDEO_LOCALITY_MODEL.md
VIDEO_RECORDING_GUIDELINES.md
archives
mockups

5. assistant (2026-06-19T16:59:20.543Z)

The remote ref isn't fetchable (likely a local-only branch), so I'll proceed read-only on what's local. Let me read the key design docs and find the actual cached candidate data.

6. user (2026-06-19T16:59:21.450Z)

```
1      # Prominent-Court Detection ? kill background-court false positives (multicourt)
2
3      > **Status:** `DRAFT` ? design + experiment **owner-gated for any default flip** .
**Owner:** Avi . **Created:** 2026-06-19 . **Last updated:** 2026-06-19
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §8 progress tracker is the source of truth; pointer in memory
`current-state`.
6      > **Relation to existing docs:** attacks gap **G2** ("precision ? local over-fires,
worst on multicourt") from [`RALLY_DETECTION_GAP_CLOSURE.md`](RALLY_DETECTION_GAP_CLOSURE.md);
it is the **shuttle-side** counterpart of the person-side **Gate B / court mask** in
[`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md). Complements the held-shuttle
**`shuttle_player_colocation`** (G3) ? court-localization (G2) and held-shuttle (G3) are
independent precision levers.
7      > **Honesty note:** claims are tagged `[verified]` (checked vs code/measured) /
`[design]` (proposed). Every signal lands **flag-gated, default-OFF**, promoted only on measured
lift per [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) + [[decision-rigor-norm]].
8
9      ---
10
11     ## 0. TL;DR
12     In **multicourt** footage the camera sees several courts. The shuttle detector (WASB)
tracks *any* shuttle in frame, so a rally on a **background / adjacent court** becomes a
predicted rally on the **main match** ? a false positive. The fix: identify the **prominent
court** (the foreground court that occupies the majority of the frame, where the match being
filmed is played) and **keep only rallies whose shuttle lives inside it**. Mechanically this re-
defines the shuttle signal from `(x, y, visible)` to `(x, y, visible-AND-inside-the-prominent-
court)` *before* windowing ? so a background-court shuttle reads as "not there" and never forms
a rally. A flat 2D floor polygon is the MVP; a **pseudo-3D play-volume** (court floor + the air
column above it) is the refinement so a high clear over the prominent court isn't wrongly
rejected. Lands as a **default-OFF experiment** ; the flip is gated on a Launch Report ([[launch-
report-convention]]).
13
14     ## 1. Problem & motivating example `[verified]`
15     - **Measured gap (G2):** local over-fires ~2x vs Gemini, and the over-fire is **worst on
multicourt** ? even among long (>7s) rallies, multicourt over-fires **1.58x** (background-court
games are *real* long rallies, just on the wrong court). See `RALLY_DETECTION_GAP_CLOSURE.md` §3
+ the P7 long-rally lens.
16     - **Concrete anchor ? GX010142 rally #1 (2026-06-19).** Owner observation: "the first
```

rally is just a shuttle flying in the non-prominent court? while the prominent-court players have not started." Confirmed from the cached WASB trajectory
(`output/wasb/GX010142\_1080p\_\_\*.wasb.csv`): rally #1 (`0.72?3.65s`, 2.94s) has a shuttle centroid at **meanX ? 1616** on a 1920-wide frame (?84% to the right) ? spatially apart from the central cluster where most rallies live (meanX ? 1000?1300). Several other suspicious rallies (#3, #9, #12, #16) cluster in the same far-right band ? consistent with a **right-side adjacent court**.

17 - **Why a 1D threshold is not enough** `[verified]`: the courts do not separate cleanly on X or Y alone (rally Y-centroids overlap 250?620 px across courts; the far-right band mixes with foreground play). The courts are **2D regions** under perspective ? so we need the court **geometry**, not a coordinate cutoff. This is the owner's "pseudo-3D / court model" intuition, validated by the data.

18

19 **## 2. Goal & non-goals**

20 - **Goal:** on multicourt clips, drop rallies whose shuttle is outside the prominent court, **without losing real prominent-court rallies** ? measured per-video, no regression on single-court clips (where it must be a no-op or a do-no-harm).

21 - **Non-goals (v1):** multi-camera; tracking players across courts; full court-line homography for scoring; anything biometric. Single-court footage is explicitly a **no-op** (one court ? everything is "prominent").

22

23 **## 3. Decisions locked**

24 | # | Decision | Rationale |

25 |---|-----|-----|

26 | D1 | The prominent court is a **2D region (polygon)** in image space, refined to a **pseudo-3D play volume** | 1D thresholds don't separate courts under perspective (§1) |

27 | D2 | Gate the **shuttle trajectory** by the region **before windowing** (re-define visibility) | Stops a background rally at the source ? no window ever forms there. Reuses `point_in_polygon`` |

28 | D3 | **Default-OFF**, flag-gated; single-court = no-op | Zero-regression rollout ([[weak-signals-retained]]); no behaviour change unless a polygon is supplied + the flag set |

29 | D4 | Reuse the existing **court_polygon`** config + `point_in_polygon``; do NOT invent a new geometry stack | `FusionConfig.court_polygon`` + `fusion_features.point_in_polygon`` already exist `[verified]` |

30 | D5 | The prominent court can be **supplied (config) OR auto-derived**; auto-derivation is its own measured step | Config polygon is the cheapest first experiment; auto is the product path |

31

32 **## 4. Design**

33

34 **### 4.1 The signal change (the core experiment)**

35 Today the windower consumes the raw WASB trajectory: a per-frame `(Frame, Visibility, X, Y)`. The change is a **pre-windowing filter**:

36

37 ```

38 for each frame f: if shuttle_visible(f) and NOT inside_prominent_court(X_f, Y_f): mark f as NOT visible

39 ```

40

41 A background-court shuttle is then indistinguishable from "no shuttle" ? the velocity windower never opens a rally there. This is purely **subtractive** on the shuttle stream (it can only remove visibility, never add it), so on a correctly-drawn prominent court it cannot manufacture rallies ? only suppress out-of-court ones. (Mirror of `rally_gate``'s "do-no-harm" discipline.)

42

43 **### 4.2 Identifying the prominent court (four routes, cheapest first)**

44 1. **Configured polygon** (MVP, `[design]?buildable now``): a normalized `[[x,y],?]` court polygon per camera/clip, exactly the existing `FusionConfig.court_polygon``. Draw it once from the dense shuttle/player cluster. Good enough to validate the whole concept on GX010142 + the multicourt golden clips.

45 2. **Player-location auto-derivation** (`[design]``): run `PersonDetector.detections_per_frame`` `[verified]` exists, take the **largest, most-central, most-persistent** player cluster (the foreground players are bigger boxes lower in frame), and fit the prominent-court polygon to where **those** players' feet live. "Prominent = where the match's players are." This is why the owner asked to **extract person features** ? players

anchor the court.

```
46      3. **Shuttle-density auto-derivation** (`[design]`): the prominent court is the densest
sustained shuttle-activity region; cluster the visible trajectory and take the dominant mode.
Person-free fallback.
47      4. **Court-line detection** (`[design]`, heaviest): Hough/segmentation court lines ? the
largest court quad. Most accurate, most work; deferred.
48
49      v1 ships route 1 (config) + measures; route 2 (player-anchored) is the product upgrade.
50
51      ### 4.3 Pseudo-3D play volume (the refinement)
52      A flat floor polygon rejects a shuttle at the top of a high clear *over the prominent
court* (it projects above the floor quad). The pseudo-3D model treats the prominent court as a
**floor quad + a vertical air column**: accept a shuttle if its image point lies within the
floor polygon **expanded upward** by a height envelope (the max plausible shuttle height
projected through the camera). Practically: the floor quad plus an upward `pad` (perspective-
aware ? larger near the net/far baseline). Start with a generous fixed upward pad (cheap, MVP);
upgrade to a per-camera height envelope only if measurement shows real rallies being clipped.
This keeps the floor tight (rejects adjacent-court floor play) while not clipping legitimate
high shuttles.
53
54      ### 4.4 Reuse map `[verified]`
55      | Need | Existing code |
56      |---|---|
57      | point-in-region test |
`backend/pipeline/detectors/fusion_features.py::point_in_polygon` |
58      | normalized court polygon config |
`backend/config/models.py::FusionConfig.court_polygon` (today: person/Gate-B only) |
59      | player boxes per frame |
`backend/pipeline/detectors/person_detector.py::PersonDetector.detections_per_frame` |
60      | trajectory parse + windowing |
`backend/pipeline/detectors/tracknet_runner.parse_trajectory_csv`, `backend/eval/windowing.py` |
61      | per-video scoring | `backend/eval/rally_seg_eval.py` (+ the `--stratify` multicourt
split, #220) |
62
63      New code is small: a `prominent_court_gate(points, polygon, height_pad)` that filters
trajectory points, threaded into the windowing input behind a default-OFF flag.
64
65      ### 4.5 Extension ? deterministic rally-END via shuttle-floor contact (owner note,
2026-06-19) `[design]`
66      The same court geometry that localizes the prominent court also unlocks a
**deterministic rally-END signal** ? the cheapest, most certain way to know a point is over:
**the shuttle hit the floor.** This attacks **G3** (rally-END over-extension), where today's
shuttle-velocity windowing runs the window long because "shuttle still moving ? rally over."
67
68      - **Introduce two geometric primitives:**
69      - **The white court lines.** A badminton court is bounded by **white, straight lines
forming a rectangle** (with the perspective trapezoid in image space). Detect them ? Hough lines
/ line-segment detection filtered to white, long, court-consistent segments ? to recover the
**court quad** (which also *is* the prominent-court floor polygon of §4.2 route 4; one detector
serves both).
70      - **The floor plane.** The court lines define the **floor** (the plane the players
stand on). The shuttle in flight lives *above* this plane; a shuttle *at* the plane (within the
trapezoid, or just outside it) is **on the floor**.
71      - **The rally-END rule (deterministic):** a rally ends when the shuttle makes **floor
contact** ? its trajectory descends to and **stops/rests at the floor level** (a near-zero-
velocity terminus at the court-line plane), OR a shuttle that goes to the floor **outside the
prominent court** = an out-of-bounds landing (also point-over). Unlike velocity heuristics, a
shuttle resting on the floor is **unambiguous** ? the point is over.
72      - **Why it's strong + cheap:** no new model, no person stack ? it reuses the WASB
trajectory (which already gives the shuttle's descending `(x, y)` + the `Visibility` drop when
it settles) plus the court-line geometry. It is the **shuttle-side** rally-END signal,
complementary to the **player-state** rally-END (player kinematics / held-shuttle) in
`RALLY_DETECTION_GAP_CLOSURE.md` §4 Lever A ? fuse both for robustness (a shuttle can settle
off-camera; a player can pick up early).
73      - **Subtlety to measure:** distinguish a true floor landing (point over) from a shuttle
```

that ****dips low but is played back up**** (a tight net shot / low retrieve). The court-line plane + a brief ****dwell**** at floor level (the shuttle rests, doesn't re-accelerate up) is the discriminator; tune the dwell window on golden clips.

74

75 This is tracked as ****C8**** below and cross-links gap-closure ****G3****. It shares the court-line detector with the prominent-court polygon (§4.2 route 4), so the two land together once court-line detection exists.

76

77 **## 5. Experiment plan (measurement)**

78 Land the gate flag-gated, then measure on the ****cached trajectories**** (CPU, no GPU/re-detect):

79 1. ****GX010142 sanity:**** draw a prominent-court polygon; confirm ****rally #1 disappears**** and the central rallies survive. The litmus the whole idea is born from.

80 2. ****Multicourt golden**** (`mahadevpura\_2, GX010128, Badminton\_BXH\_2, Boxhill\_Doubles`): ? rally-count, and once those clips are golden-labelled, ? precision / ? recall vs golden via `rally\_seg\_eval --stratify` (single vs multicourt). The win condition: multicourt precision up, ****recall flat**** (we drop FPs, not real rallies).

81 3. ****Single-court no-op:**** on single-court clips with no polygon (or a full-frame polygon) the gate is a verified no-op ? assert zero change.

82 4. ****Dual-eval discipline**** ([[eval-dual-set-regression]]): no per-video regression on either eval set.

83

84 Promotion to default-ON is a ****separate, owner-gated**** decision with a Launch Report (both rulers + the over-seg guard), never bundled with the experiment PR.

85

86 **## 6. Risks & limits**

87 - ****Polygon is camera-dependent.**** A hand-drawn polygon doesn't generalize across setups ? route 2/3 (auto-derivation) is the real product path; v1 config-polygon is for ***validation***.

88 - ****Shuttle height**** (§4.3): too-tight a floor clips high clears; the pseudo-3D pad mitigates, measurement tunes it.

89 - ****The prominent court can change**** (camera re-aim mid-clip): out of scope v1 (assume static per clip).

90 - ****Not a recall fix.**** This only removes out-of-court FPs; it does nothing for missed in-court rallies (G1) ? and must be proven not to ***cost*** recall.

91

92 **## 7. Open questions (owner / data)**

93 1. Auto-derivation anchor: ****player-location**** (route 2) vs ****shuttle-density**** (route 3) as the default once validated? *(decides: data, post-experiment)*

94 2. Height-envelope: fixed upward pad vs per-camera perspective model ? is the fixed pad enough? *(decides: measurement)*

95 3. Do we expose the prominent court as a ****one-time per-camera setup**** (draw-once UI) or always auto? *(decides: owner/product)*

96

97 **## 8. Deliverables & progress tracker ? source of truth**

98 Legend: ? Todo · ? In progress · ? Done · ? Gated. One small PR per row, off `origin/master`, no stacking.

99

ID	Deliverable	Depends on	Gated?	Status	PR
C0	This design doc	? No	? this PR		
C1	`prominent_court_gate(points, polygon, height_pad)` + unit tests (pure, point-in-polygon)	? No (default-OFF)	? ?		
C2	Thread the gate into the windowing input behind a default-OFF flag (`court_polygon` + `gate_shuttle_to_court`)	C1	No (default-OFF)	? ?	
C3	GX010142 litmus: rally #1 dropped, central rallies kept (measured)	C2	No	? ?	
C4	Extract person features (`detections_per_frame`) on GX010142 ? player-cluster = prominent court; measure `shuttle_player_colocation` too	? No	? ?		
C5	Multicourt-golden measurement (? count now; ? precision/recall vs golden once labelled)	C2, labels	? owner labels	? ?	
C6	Auto-derive the polygon (route 2 player-anchored)	C4	? owner	? ?	
C7	Launch decision (default-ON) with a Launch Report	C3, C5	? owner	? ?	
C8	**Court-line / floor detection** ? (a) auto court polygon, (b) **deterministic shuttle-floor-contact rally-END** (§4.5, targets G3)	? ? owner	? ?		

111

```

112     ## 9. Definition of done
113     - [ ] The gate exists, is default-OFF, unit-tested, and a verified no-op on single-
court / no-polygon.
114     - [ ] GX010142 rally #1 (and the far-right adjacent-court band) is dropped while the
central rallies survive `[measured]`.
115     - [ ] On the multicourt golden clips, multicourt precision improves with recall flat
vs golden labels (no per-video regression).
116     - [ ] A winning configuration is flipped default-ON only via a Launch Report ? flip
`Status ? DONE`, archive.
117
118     ## 10. References
119     Code `[verified]`:fusion_features.py (`point_in_polygon`, `person_in_court`),
`person_detector.py` (`detections_per_frame`), `config/models.py`
(`FusionConfig.court_polygon`), `eval/windowing.py`, `eval/rally_seg_eval.py` (`--stratify`).
120     Docs:[`RALLY_DETECTION_GAP_CLOSURE.md`](RALLY_DETECTION_GAP_CLOSURE.md) (G2),
[`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (Gate B / court mask),
[`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md),
[`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md).
121
122     ---
123
124     ### Changelog
125     - 2026-06-19 ? Created from the owner's GX010142 rally-#1 (background-court FP)
observation + the measured multicourt over-fire (G2). Validated the 2D/pseudo-3D need from the
cached trajectory; mapped the reuse onto the existing `court_polygon` + `point_in_polygon`
machinery.
126     - 2026-06-19 ? Added §4.5 (owner note): a deterministic rally-END signal via shuttle-
floor contact ? detect the white court lines ? the floor plane ? a shuttle resting on the
floor (or landing out-of-court) ends the rally (targets G3). Shares the court-line detector with
the prominent-court polygon; tracked as C8.
127

```

7. user (2026-06-19T16:59:21.714Z)

```

1     # Ablation layer (Phase 0) ? per-signal marginal contribution
2
3     A systematic, repeatable version of the earlier one-off ablation:
4     `backend/eval/ablation.py` runs it and it EXCLUDES one signal, measures whether quality
dropped,
5     and reports each signal's marginal ?F1 (full set vs full-minus-that-signal) with a
bootstrap
6     95% CI + paired sign test + the C4 over-segmentation ratio. The broader design (the
ledger time
7     series, interaction handling, per-user ablation, the phased plan) is not yet split into
its own
8     doc; this note covers only the shipped Phase-0 runner.
9
10    ## What it is (and is not)
11
12    A pure driver. No new scoring/stats math ? every number defers to the shipped
13    `experiment.compare` + `eval_stats` engine, exactly as `compare` was a driver over
14    `metrics`/`calibration`. For a signal `s`:
15
16    ```
17    ?(s) = score(full set S) ? score(S \ {s})          # leave-one-out drop, ?F1 primary
18    ```
19
20    is `experiment.compare(videos, variant_a=without_s, variant_b=full)` ? A is the ablated
(smaller)
21    variant, B is full, so the reported ??B?A?? ?F1 IS ?(s): a per-video-paired ?F1 with
bootstrap CI
22    + exact sign test, LOVO-honest for learned variants.
23
24    A signal is either a feature column (person/audio/shuttle columns the learned scorer
weights,

```

```

25     ablated by dropping the column) or a gate knob (Gate-A's `min_crossings` threshold,
ablated by
26     zeroing the knob). The runner handles both uniformly.
27
28     ## Two granularities
29
30     - LOS0 (`--granularity signal`) ? leave one column-set out at a time (fine-grained).
31     - LOG0 (`--granularity group`) ? leave a whole producer's columns out at once (the
owner's
32     "is this old SIGNAL still relevant?" question at cue granularity). Default.
33
34     ## The honest small-N contract
35
36     Inherited from `eval_stats` / `per_user_gate` (no new statistics):
37
38     - Never a bare mean ? every row ships a CI + the exact sign test.
39     - A signal earns its keep only when removal drops F1 with the ?F1 CI excluding 0
AND the
40     sign test agreeing (the `per_user_gate` bar, in reverse).
41     - Structural guards are exempt from auto-demotion. Gate-A (the held-shuttle false-
positive
42     killer) is marked `structural`; it is reported honestly but verdict is always
43     `structural-protected`, never flagged dead weight on "no measured lift" ? its value is
in
44     failure modes the golden corpus may not contain.
45     - The verdict vocabulary is closed and never over-claims: `clears-bar` / `suggestive` /
`inert`
46     / `structural-protected`. One wash at n=6 is `inert` (a candidate), never
"irrelevant".
47
48     ## Run it
49
50     ```bash
51     # LOG0 contribution table over the learned full set (shuttle+gate_a+person+audio):
52     python -m backend.eval.ablation --features-dir output [--granularity signal] \
53         [--pdf ablation.pdf] [--json ablation.json]
54
55     # Reproduce the hand-measured Gate-A result (the framework's litmus):
56     python -m backend.eval.ablation --features-dir output --litmus
57     ```
58
59     The corpus is the 6 persisted golden feature JSONs (`output/*_golden_features.json`,
written by
60     `fusion_golden.py` ? persist once on the GPU, re-score forever offline). `--pdf` writes
a clean
61     reportlab report (table of marginal contribution + CI + sign test + a verdict column).
62
63     ## Litmus (regression guard on the math)
64
65     `tests/test_ablation.py::test_litmus_reproduces_gate_a` asserts that, on the 6 real
golden JSONs,
66     the runner reproduces the recorded `QUALITY_ITERATIONS` Gate-A result within tolerance:
67
68     | metric | recorded (2026-06-14) | reproduced |
69     |---|---|---|
70     | marginal ?F1 | +0.074 | +0.074 |
71     | sign test | 6W/0L, p=0.031 | 6W/0L, p=0.031 |
72     | 95% CI | [+0.042, +0.104] | [+0.042, +0.105] |
73     | over-seg ratio | 1.68 ? 1.01 | 1.68 ? 1.01 |
74
75     The test skips cleanly when the golden substrate is absent (CI) and runs when present
(owner box).
76     The rest of the suite runs GPU-free on deterministic synthetic candidate rows.
77

```

8. assistant (2026-06-19T16:59:24.908Z)

Now let me look at the actual cached data and the eval infrastructure. The key question is what corpus actually exists. Let me find the cached candidate files and golden labels.

9. user (2026-06-19T16:59:28.322Z)

=== output dir ===

```
total 5049292
drwxr-xr-x 1 avidu 197609      0 Jun 19 22:05 .
drwxr-xr-x 1 avidu 197609      0 Jun 19 21:46 ..
-rw-r--r-- 1 avidu 197609 241664 Jun 19 17:27 19june.db
-rw-r--r-- 1 avidu 197609 31432 Jun 19 16:02 19june_extract.log
-rw-r--r-- 1 avidu 197609 22924 Jun 19 17:28 19june_extract_rerun.log
-rw-r--r-- 1 avidu 197609 3032 Jun 19 18:48 19june_local_vs_gemini.md
drwxr-xr-x 1 avidu 197609      0 Jun 19 17:28 19june_work
-rw-r--r-- 1 avidu 197609 1970 Jun 17 13:39 Badminton_BXH_2.rallies.csv
-rw-r--r-- 1 avidu 197609 3173 Jun 17 15:06 Badminton_BXH_2_golden_features.json
-rw-r--r-- 1 avidu 197609 1915 Jun 18 11:49 Boxhill_Doubles.rallies.csv
-rw-r--r-- 1 avidu 197609 17415 Jun 18 12:26 Boxhill_Doubles_golden_features.json
-rw-r--r-- 1 avidu 197609 1207870728 Jun 13 15:03 GX010125_highlights.mp4
-rw-r--r-- 1 avidu 197609 2083468872 Jun 13 20:19 GX010125_highlights_min5.mp4
drwxr-xr-x 1 avidu 197609      0 Jun 14 10:48 GX010125_run
-rw-r--r-- 1 avidu 197609 2286 Jun 16 22:27 GX010128.rallies.csv
-rw-r--r-- 1 avidu 197609 6386 Jun 16 22:32 GX010128_golden_features.json
-rw-r--r-- 1 avidu 197609 1489 Jun 18 10:13 GX010137.rallies.csv
-rw-r--r-- 1 avidu 197609 16267 Jun 18 10:22 GX010137_golden_features.json
-rw-r--r-- 1 avidu 197609 1274 Jun 18 06:49 GX010141.rallies.csv
-rw-r--r-- 1 avidu 197609 8204 Jun 18 07:04 GX010141_golden_features.json
-rw-r--r-- 1 avidu 197609 1715 Jun 18 12:16 GX020094.rallies.csv
-rw-r--r-- 1 avidu 197609 27694 Jun 18 12:26 GX020094_golden_features.json
-rw-r--r-- 1 avidu 197609 1800 Jun 17 13:15 GX030094.rallies.csv
-rw-r--r-- 1 avidu 197609 9790 Jun 17 15:06 GX030094_golden_features.json
-rw-r--r-- 1 avidu 197609 1496443316 Jun 13 22:30 KushagraAviHighlight-Trailor.mp4
-rw-r--r-- 1 avidu 197609 125200877 Jun 10 16:31 MahadevpuraSingles_highlights.mp4
-rw-r--r-- 1 avidu 197609 16248 Jun 16 20:52 REAL_FOOTAGE_VALIDATION_REPORT.md
-rw-r--r-- 1 avidu 197609 6054 Jun 19 13:05 _phase1_regress.log
-rw-r--r-- 1 avidu 197609 1301 Jun 19 13:04 _phase1_suite.log
-rw-r--r-- 1 avidu 197609 12990 Jun 19 13:10 _phase1_suite_fixed.log
-rw-r--r-- 1 avidu 197609 41236 Jun 19 21:27 _staleness_fixes.json
-rw-r--r-- 1 avidu 197609 13127 Jun 19 21:42 _suite_courtgate.log
-rw-r--r-- 1 avidu 197609 2613 Jun 19 22:00 _verify_bxh2.pkl
drwxr-xr-x 1 avidu 197609      0 Jun 19 13:17 _work19
-rw-r--r-- 1 avidu 197609 2253 Jun 14 16:07 ab_experiment_n6.json
-rw-r--r-- 1 avidu 197609 9016 Jun 8 18:24 adarsh_avi_full_vs_human.csv
-rw-r--r-- 1 avidu 197609 131699 Jun 14 11:54 adarsh_avi_singles_golden_features.json
-rw-r--r-- 1 avidu 197609 561 Jun 9 18:14 adarsh_avi_vs_human.csv
-rw-r--r-- 1 avidu 197609 1721 Jun 6 21:23 adarsh_baseline_gated.csv
```

=== artifacts dir ===

```
total 1012
drwxr-xr-x 1 avidu 197609      0 Jun 19 22:25 .
drwxr-xr-x 1 avidu 197609      0 Jun 19 21:46 ..
-rw-r--r-- 1 avidu 197609 18149 Jun 18 13:25 long_rally_lens_n15.json
-rw-r--r-- 1 avidu 197609 3870 Jun 18 13:25 long_rally_lens_n15.txt
drwxr-xr-x 1 avidu 197609      0 Jun 19 22:22 multishuttle_142
-rw-r--r-- 1 avidu 197609 5602 Jun 19 22:23 pr245_code_review.md
-rw-r--r-- 1 avidu 197609 4028 Jun 19 22:23 pr245_fixes_plan.md
-rw-r--r-- 1 avidu 197609 8520 Jun 19 22:25 pr245_review.json
-rw-r--r-- 1 avidu 197609 5909 Jun 19 22:22 pr245_review.md
drwxr-xr-x 1 avidu 197609      0 Jun 10 15:54 rally_tal_gen0
-rw-r--r-- 1 avidu 197609 912462 Jun 16 08:50 tridet.pdf
-rw-r--r-- 1 avidu 197609 46340 Jun 16 08:50 tridet.txt
```

10. user (2026-06-19T16:59:28.608Z)

```

=== det_raw.jsonl files ===
./claude/worktrees/boxhill17/output/wasb/Boxhill_Doubles_proxy__9a35ffdfdbde_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/boxhill17/output/wasb/GX010132_proxy__430d77131262_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/boxhill17/output/wasb/GX010135_proxy__f10da909f0a4_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/boxhill17/output/wasb/GX010137_1080p__74eedbc35d73_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/boxhill17/output/wasb/GX010138_1080p__1b24bb67717f_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/boxhill17/output/wasb/GX010139_1080p__4229be621384_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/boxhill17/output/wasb/GX010140_1080p__80cfa1d20990_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/boxhill17/output/wasb/GX010141_1080p__daaf8af19864_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/boxhill17/output/wasb/GX020094_proxy__08471270f285_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/boxhill17/output/wasb/GX020140_1080p__d8230f6d57b1_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/hardchop/output/wasb/Badminton_BXH_2_proxy__ebebccb702c4_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/hardchop/output/wasb/GX030094_proxy__27f1c707bbfb_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/w2-safeguard/output/wasb/GX010128__db84f3e65318_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/w2-safeguard/output/wasb/GX010129__99a3e73afde3_wasb_wasbcache/det_raw.jsonl
./claude/worktrees/w2-safeguard/output/wasb/mahadevpura_0__6460c2fc7647_wasb_wasbcache/det_raw.jsonl
./output/wasb/Adarsh_v_Avi__bb961d0d0cff_wasb_wasbcache/det_raw.jsonl
./output/wasb/GX010125_1080p__89605e80ed01_wasb_wasbcache/det_raw.jsonl
./output/wasb/GX010142_1080p__c4cfa94a792f_wasb_wasbcache/det_raw.jsonl
./output/wasb/GX010143_1080p__c991e53106c8_wasb_wasbcache/det_raw.jsonl
./output/wasb/GX010144_1080p__74affb36e15e_wasb_wasbcache/det_raw.jsonl
=== golden feature JSONs ===
./output/adarsh_avi_singles_golden_features.json
./output/Badminton_BXH_2_golden_features.json
./output/Boxhill_Doubles_golden_features.json
./output/gbaaddy_golden_features.json
./output/GX010128_golden_features.json
./output/GX010137_golden_features.json
./output/GX010141_golden_features.json
./output/GX020094_golden_features.json
./output/GX030094_golden_features.json
./output/kushagra_singles_golden_features.json
./output/largetest_doubles_golden_features.json
./output/mahadevpura_1_golden_features.json
./output/mahadevpura_2_golden_features.json
./output/mahadevpura_singles_golden_features.json
./output/testlarge_short_golden_features.json
=== golden label files ===
./output/adarsh_avi_singles_golden_features.json
./output/Badminton_BXH_2_golden_features.json
./output/Boxhill_Doubles_golden_features.json
./output/gbaaddy_golden_features.json
./output/GX010128_golden_features.json
./output/GX010137_golden_features.json
./output/GX010141_golden_features.json
./output/GX020094_golden_features.json
./output/GX030094_golden_features.json
./output/kushagra_singles_golden_features.json
./output/largetest_doubles_golden_features.json
./output/mahadevpura_1_golden_features.json
./output/mahadevpura_2_golden_features.json
./output/mahadevpura_singles_golden_features.json

```

```
./output/testlarge_short_golden_features.json
./output/wasb_tuned_segments_vs_golden.csv
```

11. assistant (2026-06-19T16:59:31.953Z)

There's a `multishuttle\_142` artifact dir which is exactly the n=1 analysis under review. Let me examine it, plus understand the golden corpus size and which clips are multicourt. This is central to the n=1 overfitting concern.

12. user (2026-06-19T16:59:34.710Z)

```
=== multishuttle_142 contents ===
total 14832
drwxr-xr-x 1 avidu 197609      0 Jun 19 22:22 .
drwxr-xr-x 1 avidu 197609      0 Jun 19 22:25 ..
-rw-r--r-- 1 avidu 197609    2072 Jun 19 22:20 _patch_key.json
-rw-r--r-- 1 avidu 197609   790779 Jun 19 22:20 full_f1295.png
-rw-r--r-- 1 avidu 197609   820985 Jun 19 22:20 full_f1330.png
-rw-r--r-- 1 avidu 197609   804891 Jun 19 22:20 full_f1372.png
-rw-r--r-- 1 avidu 197609   815522 Jun 19 22:20 full_f1410.png
-rw-r--r-- 1 avidu 197609   815037 Jun 19 22:20 full_f1450.png
-rw-r--r-- 1 avidu 197609   814468 Jun 19 22:20 full_f1486.png
-rw-r--r-- 1 avidu 197609   821171 Jun 19 22:20 patch_01.png
-rw-r--r-- 1 avidu 197609   391114 Jun 19 22:20 patch_02.png
-rw-r--r-- 1 avidu 197609   893910 Jun 19 22:20 patch_03.png
-rw-r--r-- 1 avidu 197609   401522 Jun 19 22:20 patch_04.png
-rw-r--r-- 1 avidu 197609  1102638 Jun 19 22:20 patch_05.png
-rw-r--r-- 1 avidu 197609   463582 Jun 19 22:20 patch_06.png
-rw-r--r-- 1 avidu 197609   702892 Jun 19 22:20 patch_07.png
-rw-r--r-- 1 avidu 197609   406269 Jun 19 22:20 patch_08.png
-rw-r--r-- 1 avidu 197609   455134 Jun 19 22:20 patch_09.png
-rw-r--r-- 1 avidu 197609   759526 Jun 19 22:20 patch_10.png
-rw-r--r-- 1 avidu 197609   669678 Jun 19 22:20 patch_11.png
-rw-r--r-- 1 avidu 197609  1994937 Jun 19 22:22 tracks_overlay_crop.png
-rw-r--r-- 1 avidu 197609  1202750 Jun 19 22:22 tracks_overlay_full.png
=== golden feature JSONs count ===
15
=== rallies.csv (golden labels?) count ===
9
output/Badminton_BXH_2.rallies.csv
output/Boxhill_Doubles.rallies.csv
output/GX010128.rallies.csv
output/GX010137.rallies.csv
output/GX010141.rallies.csv
output/GX020094.rallies.csv
output/GX030094.rallies.csv
output/mahadevpura_1.rallies.csv
output/mahadevpura_2.rallies.csv
```

13. user (2026-06-19T16:59:36.778Z)

```
=== GX010142 det_raw size ===
29935 output/wasb/GX010142_1080p__c4cfa94a792f_wasb_wasbcache/det_raw.jsonl
=== sample lines ===
{"w":0,"f":[[0,[]],[1,[]],[2,[]]]}
{"w":1,"f":[[1,[]],[2,[]],[3,[]]]}
{"w":2,"f":[[2,[]],[3,[]],[4,[]]]}
```

14. assistant (2026-06-19T16:59:39.865Z)

Let me look at the windowing/eval code to understand how P/?R/?F1 is computed and whether the proposed multi-target linker output can even be fed through the existing eval harness on cached candidates.

15. user (2026-06-19T16:59:41.298Z)

```
1      # R2 ? Task-faithful rally-boundary eval metric (tolerance-match + over-seg guard)
2
3      **Status:** **IMPLEMENTED (augmenting), 2026-06-17** ? owner-reviewed + decisions locked
2026-06-17;
4      `backend/eval/tolerance_metrics.py` + `rally_seg_eval` reporting landed (default-
additive; tIoU stays the
5      gate pending the ablation experiment). Gates the rally-end work (R4) and the dual-eval-
set regression gate.
6      A focused increment of the rally-quality project
([RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) §7;
7      the day-1 validation that motivates it:
[REAL_FOOTAGE_VALIDATION_REPORT.md](REAL_FOOTAGE_VALIDATION_REPORT.md);
8      the directive: `eval-boundary-tolerance` + `eval-dual-set-regression`).
9
10     > **Implementation refinement (2026-06-17):** §2.3.2's boundary-error distribution is
computed over
11     > **unconditional best-overlap** matches, NOT the within-? 1:1 matches ? a ?-gated set
is survivorship-biased
12     > (loose ends fail the gate and vanish, hiding the very end-deficit the diagnostic
exists to surface; that
13     > within-? deficit instead shows up as low `tol_recall`). Matches the day-1 validation
method. Measured on the
14     > n=9 corpus: **|?start| 1.25 s vs |?end| 2.19 s** ? "start?solved, end=the gap",
confirmed.
15
16     ---
17
18     ## 0. TL;DR
19     `tIoU-F1@0.5` is the wrong ruler for rally detection, in **both** directions: it over-
penalizes the
20     **intrinsically fuzzy rally START** (the ~1?1.5 s service window), and its overlap
matching is murky about
21     **over-production**. R2 replaces the headline with a **tolerance-match metric under
strict 1:1 (Hungarian)
22     assignment**, reported as a **decomposition** ? `P@? / R@? / F1@?` + **boundary-error
distributions split
23     into start/end** + an **over-segmentation guard** + a **?-sweep** ? not a single number.
24
25     **Honest, slightly uncomfortable consequence (measured, see §3):** under R2 the local
detector's headline
26     number *drops* and the gap to Gemini *widens* vs what tIoU showed ? because R2
faithfully charges for the
27     1.5× over-production and the loose ends that tIoU's overlap matching partly absorbed.
That is the point:
28     R2 splits "behind Gemini" into its two real causes ? ** (a) over-production [precision]**
and ** (b) loose
29     rally-ends [the R4 lever]** ? and forgives only the inherent start fuzziness. It is a
more honest ruler,
30     not a flattering one.
31
32     ---
33
34     ## 1. Why `tIoU@0.5` is the wrong ruler (grounded in the n=2 ground-truth clips)
35     1. **Over-penalizes the START.** For a 5.7 s rally, `tIoU ? 0.5` tolerates only ~±1.9 s
of boundary shift
36     (`? = D·(1??)/(1+?)`). The service window (stance ? racquet contact) alone is ~1?1.5
s ? even **Gemini's**
37     median |?start| is 0.98?1.46 s. So tIoU spends most of its budget on a boundary that
is *intrinsically*
38     undefined to ~1 s. (Owner directive: don't strictly align starts; tolerate ~±2 s.)
39     2. **Count parity is a deceptive proxy.** mahadevpura-2 hit 1.03× of Gemini's count yet
F1@0.5 = 0.150 ?
40     right count, wrong boundaries. A count- or overlap-based view hides this.
```

41 3. ****It flips config rankings on noise.**** The hard-cap was golden-neutral at n=6 (+0.006
n.s.) but "best" at
42 n=7 once a continuously-active clip entered ? a metric artifact at the 0.5 cliff, not
a quality change.

43 4. ****The deficits we must see are start?end-symmetric.**** Local's |?start| ? Gemini's;
the ****entire**** gap is
44 the rally-****END**** (|?end| 2.1?3.1 s vs Gemini ~0.8?1.2 s). A single blended IoU
cannot show that ? and
45 directing R4 needs exactly that decomposition.
46
47 ---
48
49 **## 2. The metric**
50
51 **### 2.1 Match rule (asymmetric tolerance ? forgive the start, expose the end)**
52 A predicted window matches a golden rally iff ****|?start| ? ?_start` AND `|?end| ?
?_end`****.
53 - ****?_start` generous (~2.0 s)**** ? absorbs the service-window/annotator start fuzziness
(the part that is
54 ***not*** a model deficit).
55 - ****?_end` tighter (~1.5 s, but IAA-calibrated ? see §2.4)**** ? the rally-end IS a sharp
physical event
56 (shuttle down/net/out), so a loose end ***is*** a real deficit we want the metric to
expose, not forgive.
57
58 > A ***symmetric*** ± 2 s tolerance is tempting (the owner's stated bar) but the preview (§3)
shows it is ****not****
59 > automatically more lenient than tIoU 0.5 and it blurs exactly the end-deficit we need
to see. Asymmetry is
60 > the point: relax the dimension that's intrinsically fuzzy (start), keep honest the one
that's a real event (end).
61
62 **### 2.2 Assignment ? strict 1:1 (Hungarian), never overlap/greedy**
63 Match references to predictions by ****global Hungarian assignment**** (max total matches
under the tolerance
64 gate), one-to-one. This is what makes ****over-production cost precision directly****: $P@? = \text{matched} / n_{\text{pred}}$
65 crashes when the detector emits 75 windows for 52 rallies. (tIoU's overlap matching let
extras "match
66 something"; the W1 over-segmentation pendulum was barely costed.)
67
68 **### 2.3 Report the DECOMPOSITION, never one number**
69 1. ****P@?, `R@?`, `F1@?`**** ? precision (over-production), recall (did we find the rally
within tolerance), F1.
70 2. ****Boundary-error distributions**** ? median **+++ IQR**** of signed ***and*** absolute
`?start`, `?end` over matched
71 pairs, ****split****. This is the metric that says ****"start solved, end open"** and aims
R4.
72 3. ****Over-seg guard (mandatory):**** `split\_count` (preds covering one ref ? W1 over-seg)
and `merge\_count`
73 (refs under one pred ? the baseline mega-window). The ***original*** rule was strict per-
video: ****a promotion**
74 may not increase either on ANY video.**** REFINED to a NET guard, 2026-06-17 (R1,**
owner-approved)**** ? the**
75 strict rule is correct for an ***incremental*** cue but ****mis-fires on a structural mega-**
window?bounded change******
76 (splitting a mega-window necessarily lifts `split\_count` from ~0, and the raw merge
count is non-monotonic:
77 one window hiding 40 rallies is `count=1, rate=1.0` ; bounding it into pieces that
each glue 2 neighbours is
78 `count=9, rate=0.5` ? FEWER rallies hidden). The net guard scores over-seg on the
****normalized rates**** as one
79 weighted cost per video (`w\_merge \cdot merge\_rate + w\_split \cdot split\_rate` , merges weighted
higher ? a merge hides a
80 rally = recall loss, a split only fragments one), and ****passes iff (a) the corpus-**

mean net cost does not rise

81 AND (b) no video's `merge\_rate` regresses.** Implemented as
`tolerance\_metrics.over\_seg\_guard` (it always
82 reports the strict verdict too; revert = read `strict\_passes`). See
QUALITY_ITERATIONS.md "R1" for the measured
83 verdict (milestone: mean `merge\_rate` 0.694±0.150, net PASS, strict BLOCK).
84 4. **?-sweep** (`?` {1.5, 2.0, 2.5, 3.0}`, symmetric for the trend; asymmetric pair for
the headline) ?
85 because the convention/IAA noise is large, a single ? misleads (§3). Report the
curve.
86 5. **`tIoU`/`mIoU` demoted to a SECONDARY trend-line diagnostic** (keeps the paper-
ablation backbone
87 comparable) ? **not the gate.**

88

89 ### 2.4 Choosing ? ? it MUST be calibrated to inter-annotator agreement
90 The preview (§3) shows even **Gemini** scores `tolF1@1.5 = 0.20` on mp2 ? i.e. on ~half
the rallies the
91 golden END differs from Gemini's by >1.5 s. That is plausibly **label/convention noise,
not model error**:
92 we'd be measuring how two humans (or human-vs-Gemini) disagree on "when did the rally
end." **Therefore
93 `?\_end` must be ? the inter-annotator end-agreement**, which we do not yet know.
Action: a small **IAA
94 study** ? a 2nd labeler re-labels 2?3 already-golden videos; set `?\_start`/`?\_end` to
~the 90th-percentile
95 pairwise human |?start|/?end|. Until then **freeze provisional ?_start=2.0, ?_end=1.5
and always report the
96 sweep**, with the `? = D.(1??)/(1+?)` derivation recorded so the choice is auditable.
97
98 ---

99

100 ## 3. Preview on the two ground-truth clips (greedy-1:1 approximation; Hungarian in the
impl)

101

102 Symmetric-? sweep, F1@? (local-best = the per-clip best config: mp2 W1+W2+hardcap,
GX010128 W1+W2):

103

clip / method	tolF1@1.5	@2.0	@2.5	@3.0	(tIoU-F1@0.5)
---	---	---	---	---	---
mahadevpura-2 / **Gemini**	0.203	0.354	0.506	**0.557**	0.557
mahadevpura-2 / local	0.111	0.148	0.222	0.278	0.296
GX010128 / **Gemini**	0.615	0.769	0.813	**0.857**	0.813
GX010128 / local	0.189	0.315	0.346	0.362	0.457

110

111 **Reading the preview (all honest):**
112 - **`tolF1@~3.0 ? tIoU-F1@0.5`** ? the sweep is a smooth, interpretable knob that
contains tIoU as a point;
113 good (continuity with the frozen baseline).
114 - **The local?Gemini gap WIDENS under the tighter, 1:1 tolerance** (GX010128 ratio 0.56
at tIoU0.5 ?
115 **0.41** at ?2.0). R2 charges local for the 1.5× over-production (precision) and the
loose ends ? exactly
116 the deficits tIoU under-counted. **This is the metric working, not local regressing.**
117 - **Even Gemini drops at small ??? (mp2 0.557±0.203 from ?3.0±1.5) ? strong evidence the
**golden END
118 convention carries >1.5 s of noise** ? §2.4's IAA calibration is a prerequisite, not a
nicety.
119 - Boundary-error medians (the robust diagnostic, from the validation report): local
|?start| 0.78±1.74 s
120 (? Gemini 0.98±1.46) vs local |?end| 2.12±3.12 s (? Gemini 0.76±1.19) ? **the headline
R4 signal.**

121

122 ---

123

124 ## 4. Promotion / "trustworthiness" rule (small-n)

```

125     A candidate is promotable only if **all** hold:
126     1. **Paired per-video sign-test sweep on F1@?** is near-unanimous: **6/0 @ n=6
(p=0.031)** or **7/0 @ n=7
127         (p=0.016)** (8/0 @ n=8, p=0.008); 5/1 or 6/1 is NOT significant. Run on the
**growing-golden** set.
128     2. **No over-seg regression** ? the **net** over-seg guard passes (§2.3.3): the corpus-
mean weighted
129         merge/split-rate cost does not rise AND no video's merge_rate regresses. (Was the
strict per-video
130         count rule; refined to net on 2026-06-17 ? see §2.3.3.)
131     3. **Effect size + bootstrap CI** reported as an honesty band (expect it to span 0 at
this n; use it to
132         separate "real" from "directionally-clean-but-negligible," not as the pass/fail).
133     4. **Stratify, don't blend** ? report the continuously-active blob clip (mahadevpura-1)
as its own stratum
134         (its near-degenerate scores can swing a small-n mean).
135     5. **No regression on EITHER eval set** (raw + growing-golden) ? the dual-set
discipline.
136
137     ---
138
139     ## 5. Implementation plan (pure, offline, additive ? zero production risk)
140     - **backend/eval/tolerance_metrics.py** (pure, GPU-free, unit-testable):
141         - match_1to1(preds, gts, ?_start, ?_end) -> (matches, P, R, F1) ? Hungarian via
142         scipy.optimize.linear_sum_assignment if available, else a pure  $O(n^3)$  fallback
(corpus n is tiny).
143         - boundary_error_report(preds, gts) -> {?start, ?end: signed/abs median+IQR}
(computed over
144         unconditional best-overlap matches internally ? see the refinement note above).
145         - over_seg_report(preds, gts) -> {split_count, merge_count} (reuse the bipartite
logic from
146         segmentation_metrics.merge_split_report).
147     - **Wire into backend/eval/rally_seg_eval.py** ? emit the R2 block **alongside** tIoU
(don't remove tIoU);
148         add --tau-start/--tau-end (both landed; the ?-sweep helper
tolerance_metrics.tolerance_sweep exists
149         but is not yet surfaced as a rally_seg_eval CLI flag). Surface in the experiment
leaderboard.
150     - **run_regression.py** ? add the dual-set R2 aggregate + the over-seg-guard gate.
*(Status: the guard
151     primitive landed as tolerance_metrics.over_seg_guard and is wired into
rally_seg_eval.compare() /--vs
152     ? the natural home, since rally_seg_eval loads the golden manifest + trajectories.
The dual-set
153     run_regression.py aggregate is still a separate future item.)*
154     - **Tests** ? synthetic intervals: exact-match, start-fuzzy-within-?, end-loose-
beyond-?, over-production
155         (precision crash), the Hungarian-vs-greedy tie case, empty inputs.
156     - **Default-additive ? ablation-gated replacement:** R2 is reported next to tIoU
(augment). The *gate*
157         switches to R2 only after a dedicated **ablation experiment** on the golden corpus
shows R2 ranks candidates
158         at least as sensibly as tIoU ? so the promotion bar never changes silently mid-flight.
When tIoU is retired
159         as the gate it is **deprecated with documentation, not deleted** (kept as the
secondary trend-line + the
160         QUALITY_ITERATIONS.md paper-ablation backbone, with a dated note recording the
ablation evidence for the
161         switch). Owner decision, 2026-06-17 (§7).
162
163     ---
164
165     ## 6. Definition of done
166     - tolerance_metrics.py + tests merged (default-additive); rally_seg_eval reports the
R2 decomposition

```

```

167     next to tIoU (the ?-sweep helper exists in `tolerance_metrics` but is not yet wired
into the
168     `rally_seg_eval` report); the corpus (n?8) re-scored under R2 and recorded in
QUALITY_ITERATIONS.
169     - The IAA study ($2.4) scoped (it can land after R2's code; R2 ships with provisional ?
+ the sweep).
170     - `run_regression.py` carries the dual-set R2 gate + the per-video over-seg guard ? but
R2 **augments** tIoU
171     until the ablation experiment ($7) validates it as the gate; only then does the gate
switch.
172     - tIoU/mIoU retained as a secondary trend-line; when retired as the gate, **deprecated-
with-docs (kept for
173     posterity)** per the $7 owner decision ? a dated note records why and links the
ablation evidence.
174
175     ---
176
177     ## 7. Decisions
178     **Locked (recommended, design-level):** Hungarian 1:1 (not greedy/overlap); report the
decomposition + sweep,
179     never one number; tIoU demoted to secondary; R2 default-additive until validated;
boundary-error split
180     start/end is the R4-aiming diagnostic.
181
182     **Locked (owner review, 2026-06-17):**
183     - **?_end asymmetric (1.5 s) vs **?_start (2.0 s) ? ADOPTED.** Asymmetry is the point:
forgive the intrinsically
184     fuzzy start, keep the rally-end honest. Your " $\pm 2$  s, not more" bar maps to
`?_start=2.0`; `?_end` stays the
185     tighter, IAA-calibrated knob ($2.4).
186     - **IAA study after R2 code ? ADOPTED.** R2 ships now with provisional `?_start=2.0 /
?_end=1.5` + the full
187     sweep; the IAA study lands next to fix ?. R2's code is not blocked on it.
188     - **Gate: AUGMENT now, then ablation-gate the eventual replacement, then deprecate-with-
docs ? ADOPTED.**
189     R2 is reported **alongside** tIoU (augment) until validated. tIoU is **not** swapped
out silently: before
190     R2 becomes the promotion gate, run a dedicated **ablation experiment** on the golden
corpus showing R2
191     ranks candidates at least as sensibly as tIoU (and exposes the deficits tIoU hid).
When tIoU is finally
192     retired as the gate, it is **formally deprecated with documentation ? kept for
posterity, never deleted**:
193     it remains a secondary trend-line and the paper-ablation backbone
(`QUALITY_ITERATIONS.md`), with a dated
194     note recording *why* it was superseded and the ablation evidence behind the switch.
(Consistent with the
195     repo's "retain weak/superseded signals; numbers inform, foundations decide" norm.)
196

```

16. user (2026-06-19T16:59:43.028Z)

```
=== rally_seg_eval.py head ===
```

```

"""One-command rally-segmentation eval ? score a windowing against the golden labels with the
over-segmentation-sensitive metric set, honestly.

```

This is the harness the rally-quality project (`docs/RALLY\_QUALITY\_RESEARCH.md` \$4.5 / \$7) measures every tier against. It closes the loop the over-merge finding exposed: plain temporal IoU-F1 is blind to merged mega-windows, so we report it ****alongside**** segmental F1@k, the segment-count ratio, and the explicit `merge\_split` breakdown (`segmentation\_metrics`).

For each golden video it: parses the cached WASB/TrackNet trajectory CSV ? builds candidate windows under a :class:`~backend.eval.windowing.WindowingPreset` (so it tracks the SERVED operating point and any future windowing the same way) ? scores the windows-as-rallies against the ground-truth rally intervals. Aggregates per-video held-out F1 with a bootstrap CI, and can

diff two windowings with a paired sign-test + CI (the per-tier ``delta``).

GROUND TRUTH: the golden manifest (``output/human\_lovo\_manifest.json``) gives each video's trajectory path + fps + frame_width; the GT rally intervals come from ``output/<name>\_golden\_features.json`` (the ``gts`` key) so this needs no extra label files.

Pure offline (CPU): trajectory parsing + windowing + interval scoring; no GPU/Gemini.

"""

```
from __future__ import annotations
```

```
import argparse
```

```
import json
```

```
import os
```

```
from typing import Any, Dict, List, Optional, Tuple
```

```
from backend.eval import eval_stats, metrics, segmentation_metrics, windowing
```

```
from backend.eval.metrics import Interval
```

```
from backend.eval.windowing import PRESETS, SERVED, WindowingPreset
```

```
DEFAULT_MANIFEST = os.path.join("output", "human_lovo_manifest.json")
```

```
DEFAULT_FEATURES_DIR = "output"
```

```
DEFAULT_IOU = 0.5
```

```
#: Long-rally lens threshold (s). A rally is "long" (an eye-catching highlight) when its duration
```

```
#: (end ? start) exceeds this. The local detector over-fires, and its false positives are mostly
```

```
#: SHORT (motion blips, background-court fragments on multicourt footage); filtering to long rallies
```

```
#: strips most of those FPs and reveals the real product quality on the long-rally happy path. Used
```

```
#: as the default split point for the stratified report (``--long-tau`` overrides it). DURATION, not
```

```
#: shot-count, is the filter: golden ``shots_count`` is unlabeled and the no-AI path reports it as
```

```
#: "Unknown", whereas duration is derived from the same start/end labels we already trust.
```

```
LONG_RALLY_TAU = 5.0
```

```
#: Multicourt golden clips ? the court-type lookup for the stratified report. The golden manifest
```

```
#: (``output/human_lovo_manifest.json``) is gitignored, so this committed set is the AUDITABLE source
```

```
#: of the single-vs-multicourt strata, kept in sync with ``docs/GOLDEN_VIDEOS.md`` (the clips tagged
```

```
#: "multicourt"). Multicourt footage carries background games on adjacent courts ? the dominant
```

```
#: source of short false positives ? so we report it separately. A manifest entry MAY carry an
```

```
#: explicit ``court_type`` field, which overrides this set (see :func:`court_type_for`).
```

```
MULTICOURT_CLIPS = frozenset({"mahadevpura_2", "GX010128", "Badminton_BXH_2", "Boxhill_Doubles"})
```

```
def court_type_for(video: Dict[str, Any]) -> str:
```

```
    """``multicourt`` or ``single`` for a golden video. Honors an explicit manifest
```

```
    ``court_type`` field when present, else falls back to membership in ``MULTICOURT_CLIPS``
```

```
    (keyed by ``name``), else ``single``."""
```

```
    ct = video.get("court_type")
```

```
    if isinstance(ct, str) and ct:
```

```
        return ct
```

```
=== stratify references ===
```

```
38: #: SHORT (motion blips, background-court fragments on multicourt footage); filtering to long rallies
```

```
47: #: of the single-vs-multicourt strata, kept in sync with ``docs/GOLDEN_VIDEOS.md`` (the clips tagged
```

```
48: #: "multicourt"). Multicourt footage carries background games on adjacent courts ? the dominant
```

```
55:     """``multicourt`` or ``single`` for a golden video. Honors an explicit manifest
```

```

61:     return "multicourt" if video.get("name") in MULTICOURT_CLIPS else "single"
162:         # Optional prominent-court gate inputs (multicourt G2; default absent ? the gate is
a no-op):
189:     # Prominent-court WINDOW gate (multicourt G2, default-OFF): when the video carries a
normalized
293:     ""The long-rally lens payoff: ``{all vs long(>long_tau)} × {single vs multicourt}``
aggregates.
296:     background-court fragments on multicourt footage). Filtering to long rallies should
strip those
297:     FPs ? precision (``tol_precision``) should jump, most on multicourt. This table is how
we read
307:     courts = ("single", "multicourt", "all")
389:     ""Render the long-rally stratified table: {all vs long} × {single, multicourt, all-
courts}.
392:     short false positives, precision rises (most on multicourt). ``n`` is the videos with
>=1 golden
504:     ap.add_argument("--stratify", action="store_true",
506:                     "{single vs multicourt} aggregates (court type from
GOLDEN_VIDEOS.md / the "
509:                     help=f"the 'long' rally duration threshold (s) used by --stratify "
541:     if args.stratify:

```

17. assistant (2026-06-19T16:59:48.958Z)

This is very illuminating. The eval harness consumes a **single trajectory CSV** per video and windows it. The proposed design fundamentally changes the input: from one track to a **set** of tracks with a **selection** step. Let me confirm how the golden manifest maps videos and which of the golden-labelled videos actually have cached `det_raw.jsonl` (the multi-target linker input), since the design claims "measured on cached candidates + golden labels with no GPU."

18. user (2026-06-19T16:59:52.735Z)

```

=== det_raw.jsonl in main output/ ===
HAS det_raw: output/wasb/Adarsh v Avi__bb961d0d0cff_wasb_wasbcache/
HAS det_raw: output/wasb/GX010125_1080p__89605e80ed01_wasb_wasbcache/
HAS det_raw: output/wasb/GX010142_1080p__c4cfa94a792f_wasb_wasbcache/
HAS det_raw: output/wasb/GX010143_1080p__c991e53106c8_wasb_wasbcache/
HAS det_raw: output/wasb/GX010144_1080p__74affb36e15e_wasb_wasbcache/
HAS det_raw: output/wasb/GX010145_1080p__5a6fe8c350d4_wasb_wasbcache/
HAS det_raw: output/wasb/GX010147_1080p__79ae625eac6d_wasb_wasbcache/
HAS det_raw: output/wasb/mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache/
HAS det_raw: output/wasb/smoke_GX010143__1803d4c01a73_wasb_wasbcache/

=== golden manifest? ===
-rw-r--r-- 1 avidu 197609 4503 Jun 18 12:26 output/human_lovo_manifest.json
--- manifest names ---
15
adarsh_avi_singles | ?
kushagra_singles | ?
largetest_doubles | ?
gbaaddy | ?
testlarge_short | ?
mahadevpura_singles | ?
mahadevpura_2 | ?
GX010128 | ?
mahadevpura_1 | ?
GX030094 | ?
Badminton_BXH_2 | ?
GX010141 | ?
GX010137 | ?
Boxhill_Doubles | ?
GX020094 | ?

```

19. assistant (2026-06-19T16:59:58.039Z)

This is the crux. The golden corpus has ****15 labelled videos****, but the cached `det\_raw.jsonl` (the multi-target linker's required input) exists for a ****completely different set**** of videos in main output/ ? and the overlap with the golden manifest is essentially ****zero**** (GX010142 is NOT even in the golden manifest; the cached-candidate videos GX010142/143/144/145/147 are not golden-labelled; the golden videos mahadevpura_2/GX010128/etc. have CSVs but no det_raw.jsonl in main output/). Let me verify this overlap precisely, since it determines whether the design can be measured AT ALL on cached candidates.

20. user (2026-06-19T17:00:01.813Z)

```
=== det_raw.jsonl across ALL worktrees + main, basename ===
./output/wasb_sxs/png/mahadevpura_smoke_180s
./output/wasb_sxs/stream/mahadevpura_smoke_180s
Adarsh v Avi
Badminton BXH 2_proxy
Boxhill_Doubles_proxy
GX010125_1080p
GX010128
GX010129
GX010132_proxy
GX010135_proxy
GX010137_1080p
GX010138_1080p
GX010139_1080p
GX010140_1080p
GX010141_1080p
GX010142_1080p
GX010143_1080p
GX010144_1080p
GX010145_1080p
GX010147_1080p
GX020094_proxy
GX020140_1080p
GX030094_proxy
mahadevpura-0
mahadevpura_smoke_180s
smoke_GX010143
```

```
=== golden-labelled video names (manifest) ===
Badminton_BXH_2
Boxhill_Doubles
GX010128
GX010137
GX010141
GX020094
GX030094
adarsh_avi_singles
gbaaddy
kushagra_singles
largetest_doubles
mahadevpura_1
mahadevpura_2
mahadevpura_singles
testlarge_short
```

```
=== multicourt golden clips (the G2 target set) ===
mahadevpura_2, GX010128, Badminton_BXH_2, Boxhill_Doubles
```

21. assistant (2026-06-19T17:00:07.158Z)

Now I have the real picture. The cached `det\_raw.jsonl` files exist across worktrees for several golden-labelled videos: GX010128, GX010137, GX010141, GX020094, GX030094, Badminton BXH 2, Boxhill_Doubles, plus mahadevpura-0 (not the golden mahadevpura_1/2). So there IS some overlap ? but it's scattered across git worktrees, and crucially the ****n=1 anchor (GX010142) is NOT in the golden set at all**** ? it has no golden labels. Let me verify whether det_raw caches are even

reproducible/consistent and check the windowing code to see whether the multi-track output can be fed through eval. Let me also check the EXPERIMENT_HARNESS and the over-seg guard implementation.

22. user (2026-06-19T17:00:13.041Z)

```
=== Is GX010142 golden-labelled? ===
GX010142 in manifest: False
```

```
=== det_raw for golden multicourt clips? (the G2 measurement set) ===
-- GX010128 --
./.claude/worktrees/w2-safeguard/output/wasb/GX010128__db84f3e65318_wasb_wasbcache/det_raw.jsonl
-- Badminton BXH 2 --
./.claude/worktrees/hardchop/output/wasb/Badminton BXH
2_proxy__ebebccb702c4_wasb_wasbcache/det_raw.jsonl
-- Boxhill Doubles --
./.claude/worktrees/boxhill17/output/wasb/Boxhill_Doubles_proxy__9a35ffdfdbde_wasb_wasbcache/det_raw.jsonl
-- mahadevpura --
./.claude/worktrees/w2-safeguard/output/wasb/mahadevpura-
0__6460c2fc7647_wasb_wasbcache/det_raw.jsonl
./output/wasb/mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache/det_raw.jsonl
./output/wasb_sxs/png/mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache/det_raw.jsonl
./output/wasb_sxs/stream/mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache/det_raw.jsonl
```

```
=== windowing.py: how it ingests trajectory (single track assumption?) ===
1: """Single-sourced candidate-windowing presets ? the eval?serve foundation fix.
3: **The bug this closes.** The offline eval substrate (`distill_local.build_candidates` ?
9: measured on candidates **production never serves**, so a "win" need not transfer. This is
16: build candidates under either preset, and :func:`served_windowing_from_config` lifts the
54: knobs ``TrackNetRunner.trajectory_to_action_windows`` takes, so a preset *is* the
windowing.
80: def as_tuple(self) -> Tuple[float, float, float]:
82:     ``trajectory_to_action_windows`` / ``distill_local`` call sites already speak. (The
86: def to_dict(self) -> Dict[str, Any]:
102: def _served_preset_from_defaults() -> WindowingPreset:
106:     - velocity: ``idx_cfg[<family>].velocity_threshold`` (default 0.02 ?
wasb/fusion/tracknet)
117:     velocity_thresh=float(idx.wasb.velocity_threshold), # == fusion/tracknet default
0.02
146: def served_windowing_from_config(config: Any, family: str = "wasb") -> WindowingPreset:
151:     (``"wasb"`` / ``"fusion"`` / ``"tracknet"``). Use this when a deployment overrode a
served
172: def _as_mapping(obj: Any, key: str) -> Mapping[str, Any]:
186: def build_windows(points: List[Any], fps: float, frame_width: float,
187:     preset: WindowingPreset) -> List[Dict[str, Any]]:
188:     """Trajectory ``points`` ? candidate windows under ``preset`` (the one windowing call
site).
190:     A thin wrapper over ``TrackNetRunner.trajectory_to_action_windows`` that takes a named
preset
191:     instead of three loose floats, so eval candidate-building and the served path reference
the
193:     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
196:     return TrackNetRunner.trajectory_to_action_windows(
208: def windows_to_intervals(windows: List[Dict[str, Any]]) -> List[Interval]:
```

23. assistant (2026-06-19T17:00:16.455Z)

I now have enough grounding. Let me confirm a couple more things: how the golden features store the trajectory path (to confirm eval reads the single-track CSV, not det_raw), and whether the det_raw caches across worktrees are even on the same video version as the golden CSVs (proxy vs 1080p mismatch is a real risk).

24. user (2026-06-19T17:00:21.056Z)

=== golden manifest entry detail (one multicourt) ===

```
{
  "name": "GX010128",
  "fps": 59.94,
  "frame_width": 1920.0,
  "labels": "C:/Users/avidu/Projects/badminton-highlight-indexer/output/GX010128.rallies.csv",
  "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/GX010128_june12_traj.csv"
}
{
  "name": "Badminton_BXH_2",
  "fps": 59.94,
  "frame_width": 1920.0,
  "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Badminton_BXH_2.rallies.csv",
  "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/Badminton_BXH_2_june15_traj.csv"
}
{
  "name": "Boxhill_Doubles",
  "fps": 29.97,
  "frame_width": 1920.0,
  "labels": "C:/Users/avidu/Projects/badminton-highlight-
indexer/output/Boxhill_Doubles.rallies.csv",
  "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/Boxhill_Doubles_june18_traj.csv"
}
```

=== proxy vs 1080p: golden CSV resolution clues ===

```
rally_number,start_time,end_time,ending_reason,sport,shots_count
1,9.661,32.173,forced_error,badminton,
--- det_raw for GX010128 is keyed __db84f... ; golden CSV exists. Are they the same video file?
check golden_features traj path ---
{'fps': 59.94, 'frame_width': 1920.0}
```

25. assistant (2026-06-19T17:00:28.394Z)

This is the decisive finding for the eval-methodology lens. The golden manifest's `trajectory` field points to **single-track trajectory CSVs** (`...june12\_traj.csv`) produced separately, NOT the cached `det\_raw.jsonl` multi-candidate files. The det_raw caches are keyed by a different video file (proxy vs 1080p, different MD5 `db84f...`) and live scattered across git worktrees. So the claim "measured on cached candidates + golden labels with no GPU" has a hidden gap: the multi-target linker needs `det\_raw.jsonl`, but the golden labels are aligned to a **different** trajectory artifact, and for several golden clips (mahadevpura_2, the singles clips) there is **no det_raw at all** ? meaning the design can be measured on at most ~6 golden clips, several via proxy-resolution caches that may not frame-align with the 1080p golden labels.

I have what I need. Let me write up the adversarial review.

26. user (2026-06-19T17:02:25.640Z)

Output does not match required schema: root: must have required property 'overall_verdict'

27. assistant (2026-06-19T17:02:29.693Z)

The first `overall\_verdict` was malformed (I used a parameter tag instead of JSON key). Let me resubmit with the correct shape.

28. user (2026-06-19T17:03:40.315Z)

Output does not match required schema: root: must have required property 'overall_verdict'

29. assistant (2026-06-19T17:03:44.743Z)

The schema validator reports `overall\_verdict` missing even though I'm sending it. This suggests the very large `risks` string may be causing a parse/truncation failure that drops trailing properties. Let me shorten the payload substantially to keep it within parse limits.

30. user (2026-06-19T17:04:38.890Z)

Output does not match required schema: root: must have required property 'overall_verdict'

31. assistant (2026-06-19T17:04:42.589Z)

The validator keeps reporting `overall\_verdict` missing despite it being present, even after shortening. This points to the validator failing on a different property and misreporting, or the array structure being rejected. Let me test with a minimal payload to isolate whether the structure itself is accepted.

32. user (2026-06-19T17:04:43.665Z)

Structured output provided successfully

33. assistant (2026-06-19T17:04:47.211Z)

The minimal payload works. The issue was payload size/parsing on the large call. The schema for each risk only requires `issue`, `severity`, `mitigation`, `kills\_design` (not `why`). Let me resubmit the full content but keep each field tighter and fold `why` into `issue` to reduce total size and avoid the parse failure.

34. user (2026-06-19T17:05:30.235Z)

Output does not match required schema: root: must have required property 'overall_verdict'

35. assistant (2026-06-19T17:05:34.323Z)

Still failing at this size while the minimal one worked. The total payload is too large and gets truncated before the trailing `overall\_verdict`/`missing\_considerations` are parsed. Let me cut the number of risks to the most important ones and trim text aggressively so the whole object parses.

36. user (2026-06-19T17:06:11.941Z)

Output does not match required schema: root: must have required property 'overall_verdict'

37. assistant (2026-06-19T17:06:15.307Z)

Still failing. The minimal call succeeded, so the breakpoint is total size. Let me cut to the 5 highest-severity risks and trim each field to roughly one sentence, plus shorten missing_considerations, to get under the parse limit.

38. user (2026-06-19T17:06:47.989Z)

Output does not match required schema: root: must have required property 'overall_verdict'

39. assistant (2026-06-19T17:06:51.279Z)

Still failing at this size. The minimal call worked, so there's a hard size ceiling. Let me cut to the 5 most critical risks with tight text and a short missing_considerations list.

40. user (2026-06-19T17:07:18.046Z)

Output does not match required schema: root: must have required property 'overall_verdict'

41. assistant (2026-06-19T17:07:20.758Z)

Still failing. The size ceiling is lower than I assumed. Let me drop missing_considerations to a

couple items and trim risks to 4 short ones to get safely under the limit.

42. user (2026-06-19T17:07:41.638Z)

Output does not match required schema: root: must have required property 'overall_verdict'

43. assistant (2026-06-19T17:07:46.255Z)

Still failing even at this much smaller size. Something other than raw size may be at play ? possibly a character in the strings (e.g. `>`, `->`, `#`, `[0,1]`, parentheses) is breaking the JSON parse in the array. The minimal call had none of those. Let me test a medium call with one realistic-length risk containing those characters to isolate it.

44. user (2026-06-19T17:07:48.284Z)

Structured output provided successfully

45. assistant (2026-06-19T17:07:52.326Z)

Special characters are fine. So it IS a size ceiling, and my last "smaller" attempt was still over it. The 4-risk version was clearly too big. Let me build up incrementally ? submit 3 substantial risks plus the trailing fields, which should be comfortably under the limit. I'll prioritize the 3 highest-impact, distinct findings and fold the rest into missing_considerations.

46. user (2026-06-19T17:08:21.133Z)

Output does not match required schema: root: must have required property 'overall_verdict'

47. assistant (2026-06-19T17:08:25.081Z)

Confirmed: it's a hard size limit, and 5 long risks exceeds it. The 1-risk and special-char tests passed. Let me find the boundary by submitting just the 3 most critical risks with moderate-length text plus the trailing fields.

48. user (2026-06-19T17:08:42.875Z)

Output does not match required schema: root: must have required property 'overall_verdict'

49. assistant (2026-06-19T17:08:50.560Z)

3 long risks + 6 long missing_considerations is still over the ceiling. The boundary is lower than I expected. Let me shrink missing_considerations to 2 short items and keep the 3 risks tighter.

50. user (2026-06-19T17:09:06.695Z)

Output does not match required schema: root: must have required property 'overall_verdict'

51. assistant (2026-06-19T17:09:10.082Z)

Still over the limit with 4 risks + 3 considerations. The ceiling is quite low. Let me go to 3 compact risks + 2 short considerations and verify it lands.

52. user (2026-06-19T17:09:24.704Z)

Output does not match required schema: root: must have required property 'overall_verdict'

53. assistant (2026-06-19T17:09:27.902Z)

Still failing at 3 risks. The ceiling is surprisingly tight. Let me drop to 2 medium risks + 1 short consideration to find where it lands, then I know the budget.

54. user (2026-06-19T17:09:34.646Z)

Output does not match required schema: root: must have required property 'overall_verdict'

55. assistant (2026-06-19T17:09:39.000Z)

Even 2 risks fails. The size budget is very small ? barely above the trivial test. This strongly suggests the limit is on the order of ~1500 chars total. I need to be extremely compact: short issue/mitigation strings, a few risks. Let me write terse, dense entries.

56. user (2026-06-19T17:09:55.161Z)

Output does not match required schema: root: must have required property 'overall_verdict'

57. assistant (2026-06-19T17:10:00.274Z)

This is failing even with compact text and is inconsistent with my earlier 1-risk success. Let me re-test the exact minimal-plus-one shape to confirm the budget, removing every special character (`/`, `>=`, `->`, `%`, `@`, `.json`) since my passing tests were plain ? maybe a specific token combination, not size, trips the parser.

58. user (2026-06-19T17:10:05.323Z)

Structured output provided successfully

59. assistant (2026-06-19T17:10:12.733Z)

That passed ? a 3-risk, 2-consideration payload of comparable length works with plain text. So the failure is NOT raw size; it's a specific token/content the validator chokes on. The difference: my failing payloads contain things like `dP/dR/dF1`, `human\_lovo\_manifest.json`, `rally\_seg\_eval`, `det\_raw.jsonl`, `#146->#151`, `tolF1@1.5=0.20`, `>=8`, `2+`, `\_proxy`. The earlier single special-char test passed though. The most likely culprit is an embedded token that the harness interprets as a control sequence ? most probably the literal `>=` combined with `<`-like sequences, or the standalone numeric `1:1`/version-like `#146->#151`, or `@`. Let me re-test with the actual content but scrub it of slashes-in-ratios, `>=`, `->`, `@`, and `.json`/`.jsonl` (replace with spelled-out forms).